

RESEARCH ARTICLE



'The tracking was in control of me': exploring affordances of self-tracking tools for adolescents' psychological wellbeing

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ABSTRACT

Self-tracking tools are increasingly used to promote enhanced physical health and wellbeing; however, emerging research suggests that their relationship with psychological wellbeing might be more complex. These considerations are particularly important for adolescents who may want to learn about their bodies at a time of rapid physiological change. This paper reports on the findings from online interviews with 29 adolescents in the United Kingdom to understand the affordances of self-tracking tools for psychological wellbeing. Findings indicate that adolescents' relationship with self-tracking is not static and whilst positive for some, others struggle with a loss of control. Calorie-tracking is particularly challenging for some adolescents. Researchers, designers, and other key stakeholders must consider what is at stake here for adolescents and assess how we might reconfigure the emotional affordances of self-tracking tools to promote control and balance and provide a system that is designed with and for adolescent self-trackers.

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Introduction

Self-tracking tools continue to grow rapidly in popularity (Jain et al., 2025), enabling individuals to monitor, control, and analyze vast quantities of personal health data (Ridgers et al., 2018). Examples of these tools range from specialist wearables such as the Oura ring, Apple Watch, or Fitbit devices, to applications (apps) that can be downloaded or are automatically available on smartphones such as MyFitnessPal, Strava, and Apple Fitness. Much of both the academic literature and marketing around self-tracking tools has centered on physical health and self-improvement (Crawford et al., 2015; Jin et al., 2020; Laranjo et al., 2021). Whilst there has been increasing focus on the entanglements between self-tracking practices and psychological wellbeing, from enhanced feelings of autonomy and social relatedness to increased pressure to perform and disordered eating behaviors (e.g. Etkin, 2016; Feng et al., 2021; Karapanos et al., 2016; Kelley et al., 2017), much remains unknown about these complex relationships.

Existing work suggests that physical activity can have positive effects on psychological wellbeing (e.g. Biddle & Asare, 2011; Chekroud et al., 2018; McMahon et al., 2016). For example, there is evidence that increased motivation to exercise and increased self-efficacy and autonomy around physical activity can have positive effects on psychological wellbeing (Karapanos et al., 2016). However, there is a delicate balance to be struck between promoting positive behavior change and negatively impacting psychological wellbeing (Eikey et al., 2017).

Another body of work has paid more attention to the darker side of self-tracking (Chen et al., 2024), exploring the ways in which self-tracking might be entwined with harmful behaviors such as disordered eating and body image concerns (Anderberg et al., 2025; Eikey & Reddy, 2017; Eikey et al., 2017; Gittus et al., 2020; Hahn et al., 2022; Honary et al., 2019; Kelley et al., 2017).

Self-tracking can have other implications for reduced wellbeing. For example, self-tracking tools can encourage rumination where 'anxious or negative self-attention occurs from engaging with data' (Eikey

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et al., 2021, p.601). This has been found with sleep tracking (Baron et al., 2017; Trabelsi et al., 2023) and negative experiences such as feelings of frustration, stress, and depression have also been reported in fertility tracking (Figueiredo et al., 2018) and symptom tracking for chronic conditions (Ancker et al., 2015). Further, tracking mental states and events can have adverse impacts on individuals with pre-existing mental health challenges. Examples include increased severity of depressive symptoms in individuals with bipolar disorder (Faurholt-Jepsen et al., 2015) and increased anxiety in individuals with psychosis (Lewis et al., 2020).

Wellbeing considerations are particularly important for adolescents, who have been recognized as a vulnerable population because of their reduced psychological wellbeing (Bonell et al., 2014; Chanfreau et al., 2013). Reduced psychological wellbeing can have negative impacts on many facets of adolescent life, including social relationships, academic achievement, and physical health (Kern et al., 2016).

Self-tracking tools do not sit in a vacuum but are part of the broader social context that adolescents contend with daily. Media reports and academic research have spotlighted the potential relation between social media use and interactions (including looking at others' profiles, commenting on posts, viewing influencer content) and adolescents' social and psychological wellbeing (Keles et al., 2020; Mabe et al., 2014; Padin et al., 2021; Schmuck, 2021). For example, 'fitspiration' (content intended to motivate individuals to pursue healthy lifestyles through exercise- and diet-related images and quotes) and pro-anorexia and pro-bulimia content can have negative effects on self-esteem, disordered eating behaviors and body image dissatisfaction (Boepple et al., 2016; Tiggemann & Zaccardo, 2015, Tiggemann & Zaccardo, 2018).

Most eating disorders first develop during adolescence and early adulthood (Keel et al., 2010). As Kent (2020) argues, self-tracking tools in combination with other social media can exacerbate the internalization of normative body images. Anderberg et al. (2025) systematic review found that the use of diet and fitness monitoring apps is associated with disordered eating, body image concerns, and compulsive exercise behaviors. Importantly, disordered eating symptomology was higher in young adults who use diet and fitness monitoring apps and among more frequent users compared with non-users. This is supported by studies of adolescents which suggest that self-tracking tools have been associated with an increase in maladaptive health behaviors in this age group, including disordered eating and compulsive exercise (e.g. Bratland-Sanda et al., 2022; Breton et al., 2022; Reynolds et al., 2024).

The language of affordances can be helpful when talking about self-tracking. The term 'affordance' stems from the field of ecological psychology where Gibson described affordances as 'what things furnish, for good or ill' (Gibson, 1966, p. 285). In Davis and Chouinard's (2016) framework, affordances are not fixed but rather technological tools afford uses and actions along a 'porous continuum' (Davis, 2020, p. 22) of interrelated *mechanisms*, whereby technologies can: request, demand, encourage, discourage, refuse, and allow. These mechanisms are *conditioned* by perception (knowing a tool's functions are available), dexterity (being physically and cognitively able to utilize the functions of a tool), and cultural and institutional legitimacy (conventions and codes that guide the ways in which people and tools relate to one another) (Davis & Chouinard, 2016; Davis, 2020). In short, '[t]he mechanisms of affordance specify how technologies afford, while the conditions of affordance situate technologies in context' (Davis, 2020, p. 13).

This paper draws on the language of the *mechanisms* and *conditions* of affordances (Davis & Chouinard, 2016; Davis, 2020) to better understand the psychological affordances of self-tracking technologies for adolescents. This work explores data from online semi-structured interviews with adolescents to address the research question: How do adolescents make sense of digital self-tracking affordances for their psychological wellbeing?

Methods

The data in this paper are drawn from a larger mixed-methods study of adolescents' (aged 13–18 years) engagement with self-tracking tools, comprising an online survey, online semi-structured interviews, and online co-design workshops. Age and location were the only inclusion criteria; participants had to be between the ages of 13 and 18 years and based in the United Kingdom (UK) to be eligible to participate. When completing one research activity (e.g. survey), participants could consent to be contacted about other activities (e.g. interviews and co-design workshops); however, participation in each activity was not

contingent on taking part in another activity. This paper focuses on data from the semi-structured interviews. Details of the full study are available elsewhere (Freeman, 2023) and findings from the co-design workshops have been previously published (Freeman & Curtis 2022, 2023).

Participants and procedure

This study was informed by an interpretivist paradigm, foregrounding the lived experiences of different individuals engaging in social interaction (Klein & Meyers, 1999). Semi-structured interviews were conducted between 20 July 2021 and 8 December 2021 on Microsoft Teams. These interviews took place during various phases of COVID-19 restrictions in the UK. There were three separate interview guides for current users, previous users, and non-users of self-tracking tools. There were also distinct interview guides for those under the age of 16 and those over the age of 16. The duration of the interviews differed between these two groups such that those below the age of 16 had shorter interviews (approximately 30 minutes) than those above the age of 16 (approximately 45 minutes) to account for fatigue and school scheduling. The interviews were conducted by JF. Sample interview guides are available in Appendix 2.

Participants were recruited through several channels. Study advertisements were distributed on social media and by various youth organizations, in addition to the researchers' personal networks. The sample size for these interviews was determined by pragmatic considerations (e.g. adolescents' online availability during the COVID-19 pandemic) and was in line with previous studies in this area (e.g. Lupton, 2018; Radovic et al., 2018). Parental consent and participant assent were obtained online for participants under the age of 16. For those over the age of 16, online participant consent was given. Prior to data collection, this study was granted ethics approval by Oxford University Social Sciences and Humanities Interdivisional Research Ethics Committee Approval (Reference: R75812/RE006). Participants were compensated with a £10 online shopping voucher for their time.

In total, 16 adolescents aged 13–15 years and 48 adolescents aged 16–18 years completed online consent/assent forms to participate in an interview. From there, 33 adolescents went on to schedule an interview and 29 successfully completed an online interview (see Figure 1). Of these, 12 were aged 13–15 years (41.38%) and 17 were aged 16–18 years (58.62%). In terms of gender composition, 14 identified as female (48.28%), 11 identified as male (37.93%), one identified as non-binary (3.45%), and three did not disclose their gender (10.34%). Further, 16 were using self-tracking tools at the time (55.17%), eight had previously used self-tracking tools (27.59%), and five had never used self-tracking tools (17.24%). Tools used by participants included wearables such as Fitbit and Apple Watch and apps such as MyFitnessPal and Strava. A summary table of interview participants is available in Appendix 1.

Data analysis and reporting

Interviews were video-recorded and transcribed verbatim using Microsoft Word. The transcripts were transferred to NVivo 12 (QSR International Pty Ltd., 2019) for analysis. Braun and Clarke's reflexive thematic analysis (Braun & Clarke, 2006, 2022) was used to analyze the data, recognizing researcher subjectivity as an important data analysis tool (Braun & Clarke, 2022). Reflecting on this subjectivity was essential to the research process and brief notes were kept in a research diary (Nadin & Cassell, 2006). Tracy's (2010) eight big-tent criteria were used to evaluate the quality of this qualitative study. Within this, rich rigor was demonstrated through a clear identification of key theoretical constructs guiding this research (i.e. affordances) and clear explanations of the research context and data collection and analysis processes.

JF analyzed the data as a single coder, working in line with standards and good practice in reflexive thematic analysis (Braun & Clarke, 2022), moving recursively through the six phases of the process: familiarizing oneself with the data, generating initial codes, generating (initial) themes, reviewing potential themes, defining and naming themes, and producing the report (Braun & Clarke, 2022). A combination of semantic and latent coding was used (Braun & Clarke, 2022). Transcripts were initially coded inductively and semantically, staying close to the data. When refining the themes, attention was paid to the latent

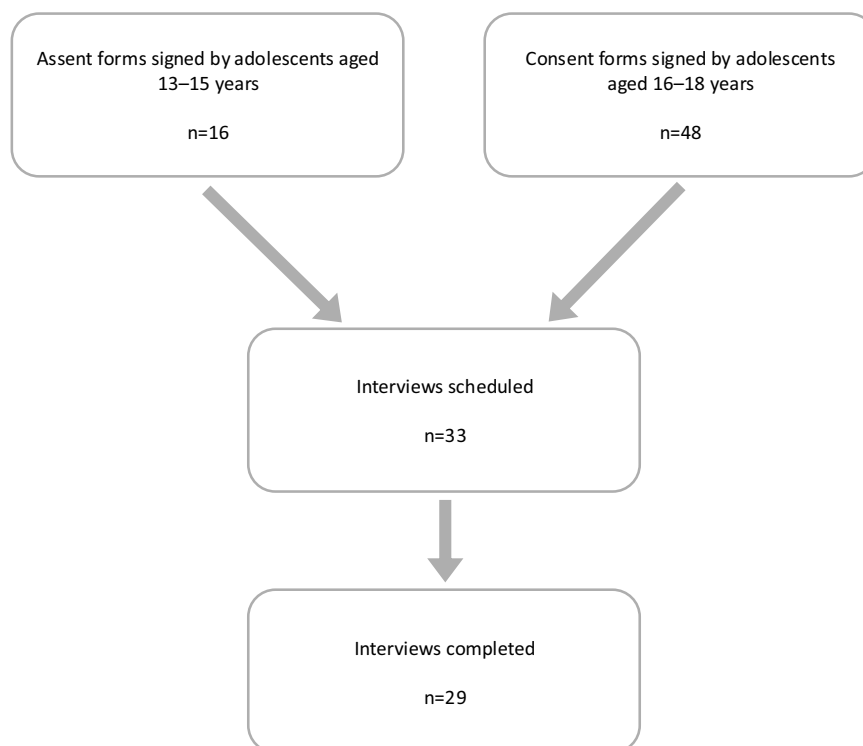


Figure 1. Flowchart of interview sign-up process.

meanings in the data, and the use of theory (namely affordances) was central to the analysis and interpretation of the data. As Merriam and Tisdell argue, ‘The sense we make of the data we collect is equally influenced by the theoretical framework. That is, our analysis and interpretation—our study’s findings—will reflect the constructs, concepts, language, models and theories that structured the study in the first place’ (2016, p.88). As such, where appropriate, relevant literature is integrated into the Results section alongside interview data.

The Results have been written to foreground adolescent voices. Each section begins with a participant vignette to illustrate the relevant theme. Quotations (attributed in the form ‘(Pseudonym, Age)’) are used extensively throughout to emphasize adolescents’ own words.

Results

The following section is structured in three key themes, titled: the numbers game; the slippery slope; and the battle for control (see Table 1).

The numbers game

Kirsty’s story

Kirsty (15) has tried different self-tracking apps but has found MyFitnessPal to be the only sustainable option for her as it keeps everything in one place and is intuitive to use. She enjoys monitoring her progress and finds that viewing her data gives her a ‘*sense of security*’. She describes herself as someone who puts herself under a lot of pressure and has previously set ‘*unrealistic goals*’ with her tracking. She is aware that her personality might predispose her to become fixated on her data and so she is careful to ensure that this does not happen to her. For example, she chooses not to share her data with others as she is worried it would be too easy to become competitive. Whilst Kirsty currently self-tracks, she is grateful she did not start earlier in life when she feels she may not have been equipped to engage in these practices positively.

Table 1. Summary of themes and sub-themes.

Theme	Sub-Themes	Definition
The numbers game	Comfort in data	This theme captures adolescents' search for reassurance through tracked data and the value they saw in tracking their bodies and health to support their physical and psychological wellbeing.
The slippery slope	No appetite for calories The self beyond the numbers	This theme reflects the darker potential of self-tracking to encourage behaviors and thoughts that were detrimental to adolescents' psychological wellbeing, particularly with respect to calorie-tracking.
	Too little is never enough	
The battle for control	When tracking takes over	This theme highlights the challenges adolescents found in maintaining control of their tracking behaviors and the strategies they implemented to reconnect with their bodies.
	Seeing the world through data	
	Taking the reins	

Comfort in data

Hemberg et al. (2024) found that for young people, the COVID–19 pandemic was marked by impacts such as changes in social networks, an increased need for support, and increased loneliness. At a time of such disruption, the search for comfort was evident throughout many of the interviews in this study. Adolescents described how self-tracking could reduce anxiety, particularly as they were managing an unfamiliar environment (e.g. lockdowns, school closures). Some adolescents shared how self-tracking their exercise allowed them to maintain their sense of identity as athletes even whilst team sports and training was restricted and this had positive effects on their wellbeing.

Some adolescents described positive experiences of how self-tracking could benefit both their physical and psychological wellbeing, highlighting motivations to begin self-tracking centered around self-improvement and -understanding. Once they started tracking, some found comfort in the data they were generating and collecting. The sense of the 'objectivity' of the data and the achievement of reaching goals were reassuring for some adolescents and, like Kirsty (15), they found a '*sense of security*' in the numbers. For example, when describing her sleep and exercise tracking, Natalie (16) shared: '*it kind of like helped with my mind [...] being able to see it [my goal] and seeing whether or not I've achieved it kind of put my mind at rest.*'

This was similar for tracking chronic illness. For example, Miriam (17), who tracked symptoms of a long-term health condition (polycystic ovary syndrome, PCOS), found that self-tracking '*just kind of helped to like put me at ease, I guess*'; rather than '*speculating*' about what might be happening with her body, she could rely on the data to tell her. Adolescents described that this improved their sense of wellbeing as they used the data to help them to achieve a sense of certainty, easing their anxieties about their bodies and health as they learned more about themselves.

Nevertheless, whilst some found comfort in their tracked data in the short term, stories of long-term positive effects on psychological wellbeing were rare. Instead, adolescents painted a more complicated picture of the relationship between their self-tracking practices and psychological wellbeing.

The slippery slope

Miriam's story

Miriam (17) uses a menstrual cycle tracking app, Clue, to manage her PCOS. She was diagnosed at the age of 12 and was offered little support from doctors so felt that she had to bear the responsibility of learning more about herself and to start tracking her health. Miriam has found that tracking has stopped her speculating about her health, provided reassurance, and allowed her to plan her life around her PCOS. Miriam has also tried using calorie trackers but found that this experience was quite detrimental to her psychological wellbeing. She set out with the intention of gaining weight but through her tracking her goal slowly shifted, and she found herself instead trying to lose weight. She also began to get anxious about eating in social settings where she could not meticulously weigh her food down to '*every single grain of*

rice'. Miriam now vocally opposes calorie-tracking behaviors and has since stopped tracking calorie data herself.

No appetite for calories

Beyond the initial excitement and novelty of self-tracking, adolescents saw the potential for tracking to take a darker turn. Adolescents who had previously self-tracked but were not currently tracking at the time of the interview more readily described these more negative experiences. Whilst none of the interview questions asked specifically about the tracking of calories, this topic came up repeatedly without prompting. Calorie-tracking was viewed as decidedly distinct from other forms of self-tracking (e.g. exercise and physical activity, sleep, symptoms) and as a type of tracking that raised several complications for adolescents' wellbeing.

Many of the adolescents were particularly concerned about the prospect of calorie-tracking, which is consistent with previous work (e.g. Potapov et al., 2021): *'I would definitely say that calories are the biggest problem. Whether it's tracking how many calories you're eating or tracking how many calories you're burning'* (Ophelia, 17). Similarly, Miriam (17) shared her concerns about *'the whole calorie thing... I think there's like a million and ten different negatives and downsides to that.'* For adolescents, calorie-tracking represented a slippery area of self-tracking that could lead to problematic relationships with food and their bodies.

In contrast to adolescents' descriptions of the motivating nature of physical health tracking, they saw calorie-tracking as more *'constricting'* (Rachael, 17). Quinn (17) described this as follows: *'if you tracked food, I feel that would take away from the experience of eating and enjoying yourself whereas if you tracked your running and stuff, I feel that motivates you more rather than takes it away.'*

Whilst adolescents often oriented their comparisons of step count around the value of 10,000 steps, for calorie-tracking there was little sense of an established 'normal': *'with steps there are 10,000 and that's kind of universally agreed on. Then I think with calories it's like, there are different opinions on each thing'* (Sawyer, 16). In the absence of these norms, many adolescents described trying to keep the number as low as possible and eat *'virtually nothing and lose weight'* (Ophelia, 17), reflecting hegemonic ideals of 'thinness' (Greene & Brownstone, 2021). Adolescents felt that there was *'a different social response'* (Sawyer, 16) around these data such that the moral imperatives and value judgements made about calorie-tracking stood out as different to other types of physical health tracking.

These moral judgements and standards were revealed in adolescents' stories of complex relationships between self-tracking and psychological wellbeing. Adolescents' descriptions of these practices were so intricately entwined with food and consumption. For example, there were clear patterns in the language and metaphors that adolescents used around self-tracking and the broader social media ecosystems embroiled in these practices. Although adolescents carefully controlled their calorie intake through self-tracking, they described being *'fed'* by social media content; for some, nutrition went digital. Conversations around *'consuming'* content and social media *'feeds'* permeated the interviews. Edward (17) described satirical Instagram workout videos as *'the only fitness content I consume'*. When adolescents had over-indulged in this online space, they often engaged in digital *'detoxes'* (a period of time away from social media and the consumption of *'unhealthy'* content); however, interestingly, self-tracking tools were largely exempt from these detox practices.

As someone without experience of self-tracking, Adaline (16) still felt that social media companies offered a generic content *'diet'* before the algorithms became more personally calibrated to unique tastes and dietary requirements. She described her experience with TikTok: *'when you download the app, the first things you see are 'Oh here's a video of someone with a kitten and some food and then a fitness video'. Those are kind of the sort of platter that they provide for you at the start'*. Overconsumption of this *'platter'* of health and fitness content could present challenges for adolescents: many responded to this indulgence by restricting their *'offline'* food diets.

Even adolescents who were more cautious with their online *'diet'* sometimes found this content creeping in: *'I don't follow any pages specifically for health and fitness, but on my feed it has quite a lot'* (Jasper, 14). It is here that the old adage *'you are what you eat'* might hold some water. Adolescents described how the content consumed online could shape their identity and sense of self, and for some, this

could contribute to harmful experiences, such as disordered eating behaviors and body image concerns. Seeing the ways in which these consumption metaphors seeped into adolescents' language about health and fitness content on social media more generally, it seemed that there was little escape. Food was a critical component of these messy relationships for many adolescents.

The self beyond the numbers

Some adolescents saw their particular stage of life as a time of unique vulnerability to the negative psychological effects of self-tracking practices. Even adolescents who had not engaged in self-tracking practices themselves shared concerns about potential harm. They were wary of the dangers of tracking for their age group:

Like the diet teas and the detoxes and stuff [...] I think these self-tracker apps are going to be the next ones which are gonna kind of come out as being we shouldn't be letting kids have access to these apps. And I use kids vaguely. up to 18. (Adaline, 16)

Miriam (17) shared that adolescents were often inclined to set unachievable goals, particularly around calorie-tracking. She imagined that this was the cause of a lot of harm:

So like with adolescents, I'm sure you know like stuff around like body image is like really, you know like a big thing I guess for us [...] when people come to counting calories they can get really obsessive after a while [...] it's like 'I want to look like this person. I want this body shape or whatever and to get there I need to get there I need to be eating a certain amount or I need to be restricting a certain amount' or something like that and it can just be really destructive with us.

This more 'destructive' (Miriam, 17) aspect of tracking was perhaps magnified by the COVID-19 pandemic and periods of lockdown, when adolescents sought comfort and became more 'attached' (Quinn, 17) to their metrics and 'thought too much about them because there was nothing else to think about' (Quinn, 17).

The link between eating disorders and self-tracking was clear to the participants. As Adaline (16) remarked: 'I don't think there's anyone I've seen on one of these social medias that has an eating disorder that hasn't used a self-tracking app at some point'. Similarly, Hannah (16) said, 'I know calories has had a big impact on like my age group in particular with umm eating disorders and stuff'. Natalie (16) shared her personal struggle with bulimia nervosa and how her self-tracking practices became entwined with challenges around food:

With my bulimia, when it came to like recording what I've eat and stuff like that, I'd think 'Oh have I eat too much?' Or if I searched it up and see how many calories I'd be like, 'Oh my gosh, no I can't have that, I can't have that'. It got to a point where like whatever I was eating, I knew the calories and I knew how much exercise to do to burn it off straight away.

The immediate availability of data was an affordance of self-tracking tools that encouraged negative emotional responses for some. Natalie (16) went on to describe how, under these circumstances, the presentation of tracking data became an external presentation of her own bodily sensations of inadequacy and failure:

Obviously being bulimic, when things don't go your way or if you're having a bad day, they do make it worse because you kind of feel like a failure when you do see that thing in front of you. Like you can already feel it within yourself but... but being able to just pick up your phone and it being there and like it's right in front of it, it does make you feel bad after a while.

For these adolescents, self-tracking was about much more than just numbers and reflected something deeper about themselves and their own experiences. Self-tracking was interwoven with challenges they faced in their environment and emerged in complex relationships between their sense of self and other concerns such as nutrition and mental health challenges.

Too little is never enough

Participants found the rigid presentation of calorie data allowed by self-tracking tools to be demotivating and expressed a preference for alternative framings: 'rather than calories burnt, make it more like positive

reinforcement and make it like how much stronger you've gotten' (Hannah, 16). Hannah's suggestion points to a curious tension in the self-tracking behaviors described by the adolescents. Where tracking physical activity was often presented as having more positive effects on psychological wellbeing, many of the adolescents described tracking aspects such as steps, distance, or weights lifted as having a focus on growth and getting 'stronger' (Hannah, 16). They strove to drive these numbers to be as high as possible.

By contrast, many of the calorie-tracking behaviors described by adolescents centered around keeping calories consumed as low as possible, and this goal was often associated with a negative impact on psychological and physical wellbeing. The directionality of the tracking goals and motivations for self-tracking therefore raised challenges for adolescents when they were striving towards a number that was as low as possible. Many adolescents spoke about the particular challenge of aspiring to have an idealized body type that is often unachievable: *'trying to reach that sort of ideal body type, which actually doesn't exist [...] you're just kind of pushing yourself until the limit of forever'* (Miriam, 17).

Like Miriam (17), many adolescent girls described a disconnect between their initial intentions when they started to track their calories and the obsessive place in which they landed through the self-tracking practices afforded by their tools. Ophelia (17) shared a similar story of striving towards this ideal and the harmful effect this had on her mental health:

I first started as I was trying to lose a bit of weight and so I was counting my calories, seeing how much I was eating during the day to see what I would need to maintain weight and to lose weight and then that ended up never being enough and so I kept cutting down and cutting down and it was so unhealthy. But you don't really realize it until it's kind of too late and you have to get help for it 'cause you can't do it on your own anymore.

The emotional affordances of seeing that they had not reached these goals were extreme—*'devastation and guilt and like worry within myself'* (Ophelia, 17)—and this could be *'overwhelming'* (Sawyer, 16). After falling down this slippery slope of calorie-tracking, desperately chasing an ideal body, many adolescents found that their relationship with both the self and tracking fundamentally changed, and they felt as if they had spiraled out of control.

The battle for control

Hannah's story

Hannah (16) self-tracks using her phone. She previously used a Fitbit but she found that this was too readily available and meant that she was checking her data too often, which she felt negatively impacted her psychological wellbeing. She discussed her step count with her colleagues and used this to build relationships and establish social connections at work. At one time, she tracked her calorie intake but found she was too preoccupied with this. She found this negatively impacted her relationship with food and she would feel *'guilty'* after eating.

When tracking takes over

In line with previous work (e.g. Freeman & Neff, 2023), *'the pressure everyday to reach the numbers'* was considered *'dangerous'* (Quinn, 17) for some adolescents as it could encourage them to become fixated on data. Many adolescents, both currently and previous self-trackers, were reflective about moments where they had crossed the threshold and lost control of their tracking: *'I definitely think I've kind of gone too far into it'* (Sawyer, 16).

Adolescents resisted the idea of continuous tracking or *'the idea that we always need to do it'* (Miriam, 17) and they worried that *'if you're really obsessive over it, it can just like backfire on you'* (Miriam, 17). They wanted to maintain choice regarding when they self-tracked and advocated the importance of having breaks from tracking to maintain a healthy relationship with quantified data: *'it's like anything, if you obsess over it just gets to a point where you need to take a break and step back from it because it's all you're thinking about'* (Morgan, 16). This aligns with Gorm and Shklovski's (2019) notion of episodic tracking whereby users freely move in and out of active tracking to best suit their needs. Adolescents flagged the dangers of continuing to self-track beyond the point at which they felt in control but described the challenge of these tools discouraging more autonomous, episodic use: *'The tracking was in control of me.'*

So yeah, I guess that's why I didn't like it 'cause it didn't really feel like I was in control, which kind of defeats the whole purpose of self-tracking' (Miriam, 17).

For some, the availability of self-tracking data to quantify their experiences discouraged their enjoyment of their usual activities, such as exercising, eating, and spending time with family and friends. Adolescents expressed concerns about what happens when *'everything kind of revolves around it [self-tracking]'* (Luke, 16) and felt that existing tools did not allow them to take meaningful steps to remain in control. They described experiences of being *'slightly too concerned about, you know, the numbers in the gym'* (Mikhail, 17).

Some adolescents were wary that this loss of boundaries could make them lose touch with their bodies: *'I think trackers can sometimes take away from that intuition and your ability to kind of read your own body'* (Jasmine, 17). This sat in tension with the core ideals underpinning self-tracking and the Quantified Self's promise of 'self-knowledge through numbers' (Quantified Self, n.d.). Ironically, gaining these personal, bodily insights made adolescents feel more alienated from themselves and their bodies. Adolescents were hesitant to become *'dependent'* (Quinn, 17) on self-tracking tools or for their bodies to be reduced to numbers because they were aware that when it comes to self-tracking, *'it doesn't have all the answers'* (Quinn, 17). Instead, adolescents were determined to remain in control of their self-tracking practices and emphasized the importance of choice regarding how they intuited their bodies' signals.

Seeing the world through data

One consequence of losing their sense of control over tracking was that adolescents began to view their worlds through the lens of data alone. The affordances of self-tracking tools could encourage everyday routines of work, walking, and eating to become obscured by a data-centric view. When tracking data moved beyond the materiality of the device it began to *'take over your life'* (Adaline, 16) such that adolescents would *'constantly be on those apps 24/7'* (Ophelia, 17) and they found that they began to prioritize viewing their daily activities through this prism:

Like when I was working and things and... and my priority would be I've got a burn this amount whilst I'm working and things like that. You know it should be about doing the best work I can [...] or like enjoying my day instead of worrying about what I'm eating and how much I'm exercising. (Ophelia, 17)

Once again, this was particularly true of calorie-tracking, which shaped the way that some adolescents viewed food and the world around them. Tracking could mask the enjoyment or experience of the activities themselves, and ordinary objects were often seen through the lens of data: *'when you go around a supermarket, you can look at this stuff and go 'Ohh that has X amount of calories' because you put it into the app enough times that you know'* (Adaline, 16), or *'every time we go to a restaurant you can't look at anything without going, 'Oh, I'd need to run 5 minutes to get rid of that''* (Adaline, 16). Adolescents described in detail how food became tied with exercise; they would pay penance for eating by being sure to burn off any calories consumed.

Unfortunately, these associations did not disappear if adolescents stopped using tracking devices or apps, and instead they were *'kind of hard to shake off'* (Miriam, 17). Like other young adults (Clark et al., 2022), adolescents described how self-tracking tools encouraged a notion of deeply internalized tracking whereby they had tracked for so long that they instinctively knew how many calories were in different foods or were burned through different activities. Ophelia (17) described this: *'it's just kind of like those numbers end up getting stuck in your head, and so when you next go to eat it, you just see those numbers instead of the food that it is'*. The quantification of food consumption was a common experience shared by many adolescents: *'I can still look at an apple and be like, 'Oh, that's so many calories'. Or like look at a piece of bread and be like 'Oh, that bread's got that many' [...] I've just learned it now'* (Ophelia, 17).

Whilst for some, internalized tracking could be a helpful shortcut and a sign of learning and growth, in such circumstances, the consequences were darker. This *'changing perspective of food'* (Sawyer, 16) encouraged by self-tracking tools was common amongst adolescent girls. The impact of this was that negative emotions were often tied with food, and adolescents described a sense of shame regarding their eating practices: *'I never got to the point where I stopped eating the packet of crisps, but it definitely... I felt guilty afterwards'* (Hannah, 16).

This data-focused lens could also have an impact on their social relationships, even when their tracking practices were private. For example, Miriam (17) shared:

I started en- enj- not enjoying things that I used to previously enjoy it like umm I used to get a lot of anxiety around like going out with my friends to eat and stuff. Because I'd feel like, 'Oh, I don't have my weighing scale on me, I can't measure what I'm going to eat' or 'The menu doesn't show the calories and stuff, and I can't put it into my tracker.'

Taking the reins

Adolescents clearly articulated approaches they had used to combat negative psychological effects of tracking and regain control. This has been shown in adults too, where users creatively resist the standard and orient towards the personal; they 'hack' self-tracking tools to collect new data (Neff & Nafus, 2016) and engage in 'episodic use' in order to ensure that the tracking still suits their needs (Gorm & Shklovski, 2019, p. 2505). For some adolescents, the fight for control centered around the materiality of self-tracking tools and putting strategies in place to limit the visibility of their—and others'—data, such as deleting apps or ceasing to use wearables: *'I've deleted apps and I've kind of got rid of some of my tracking because I. it led into unhealthy habits for me'* (Ophelia, 17). Similarly, in the story at the beginning of this theme, Hannah (16) described starting to just track her data on her phone as they were too readily available on her Fitbit:

I felt I was looking at it too much in a way, maybe. It being constantly on my wrist was sort of like I was always checking and like even just out of boredom. But I guess that sort of played a psychological effect.

Some found after they had lost their sense of control over their self-tracking, they were able to continue to track; however, they had to modify the availability of their data. For example, Hannah (16) stopped engaging in active tracking behaviors (e.g. calories) whilst retaining more passive tracking (e.g. steps). Adolescents were reflective about the nuanced meaning that different metrics had for them and used various methods to adapt their use of tracking tools. They were able to look beyond 'self-tracking' as a broad practice and instead incisively pointed to the particular types of tracking that were more problematic than others.

Moreover, control over the materiality of devices extended beyond simply ceasing to use them temporarily or permanently. Instead, adolescents experimented with structuring their data environments so that tracking captured less of their day-to-day attention. For instance, Ophelia (17) said that when she felt out of control of her tracking, her calorie count was omnipresent in her day-to-day activities: *'when I was at a point that was bad, my like my watch face was just how many calories I'd burned during the day so it was all I would see'*. Nonetheless, she devised a strategy to regain some control over her tracking: *'I changed my watch face to something that like a photo so it would be a lot nicer'* (Ophelia, 17). This demonstrates how the affordances of self-tracking tools are sociomaterially constituted and that these arrangements allowed adolescents opportunities to set personal boundaries for their engagement with these technologies and to reimagine the affordances of self-tracking tools to reassert their control.

Adolescents were aware of approaches they could use to regain command over their bodies and their data through active choices; however, some remained concerned about those who might not be as adept at implementing such strategies. They warned of certain sociomaterial arrangements that could lead someone to be disproportionately susceptible to these negative psychological affordances: *'if you are prone to kind of wanting to control that aspect of your life, then I don't think you should [self-track] because it will just lead to bad places'* (Sawyer, 16).

Some adolescents articulated that their strategy for maintaining control was to intentionally avoid engaging in self-tracking practices generally or tracking certain kinds of data more specifically: *'I've kind of like stayed away from that whole area [calorie-tracking] because I'm not gonna lie, it kind of scares me'* (Rebekah, 15). They had seen 'problematic' (Rebekah, 15) outcomes for people who 'relied on it' (Rebekah, 15) and did not want this for their own lives so they *'don't touch those'* (Quinn, 17) areas of tracking.

Discussion

The interview data in this paper offer a deep account of the lived experiences of individual adolescents. Drawing on the concepts of the *mechanisms* and *conditions* of affordances (Davis & Chouinard, 2016; Davis, 2020), the findings in this study have demonstrated that self-tracking can be a double-edged sword for this group. Adolescents recognize the potential benefits and comfort of seeing their data; however, the affordances of self-tracking tools can encourage harmful behaviors, particularly when calorie-tracking. As one participant highlighted, obsessive engagement with self-tracking tools can lead to negative outcomes for young people: *'if you're really obsessive over it, it can just like backfire on you'* (Miriam, 17).

Whilst much self-tracking advertising promises users a deeper sense of self-knowledge and bodily control through measurement (Crawford et al., 2015; Fors et al., 2020), this does not acknowledge the potential for engagement with these tools to have (unintended) negative consequences for users' well-being. This may be particularly true for adolescents who are often engaging with self-tracking tools that have been designed for adult consumers (Schaefer et al., 2016; Wartella et al., 2015), leading to frustration (Goodyear et al., 2018) or features misaligned with their needs (Wartella et al., 2015).

Whilst focused on adolescents' experiences, many of the insights from these interviews align with the findings of Chen et al.'s (2024) broader systematic review of potential harms of self-tracking tools. Negative psychological outcomes they identified included both 'negative emotional reactions' (e.g. guilt, anxiety, pressure/stress, frustration) and 'maladaptive cognitive reactions' (e.g. body image dissatisfaction, rumination) (Chen et al. 2024, p.154). They also highlighted disordered eating behaviors and compulsive exercise as negative behavioral outcomes from engagement with self-tracking tools, and underscore the potential for negative social outcomes, such as social isolation. Chen et al. emphasize the importance of further research with vulnerable groups. This underscores the urgency of understanding the potential implications of self-tracking adolescents, who are navigating a life stage of greater psychological vulnerability (Bonell et al., 2014; Chanfreau et al., 2013).

The changing relationship between adolescents' psychological wellbeing and self-tracking practices

By narratively following adolescents' journeys through self-tracking from initial excitement and comfort in data to the slippery slope of calorie-tracking and eventually to a loss (and potential regain) of control, the circuitous trajectories that self-tracking can take in relation to adolescents' wellbeing become clearer (see Figure 2). This figure was created by examining the themes drawn out of the transcripts of adolescents who had experience with self-tracking and tracing the common self-tracking paths they described in their interviews. Particular attention was paid to adolescents who had previously self-tracked and the moments at which they described choosing to stop. This figure is not intended to capture all experiences but

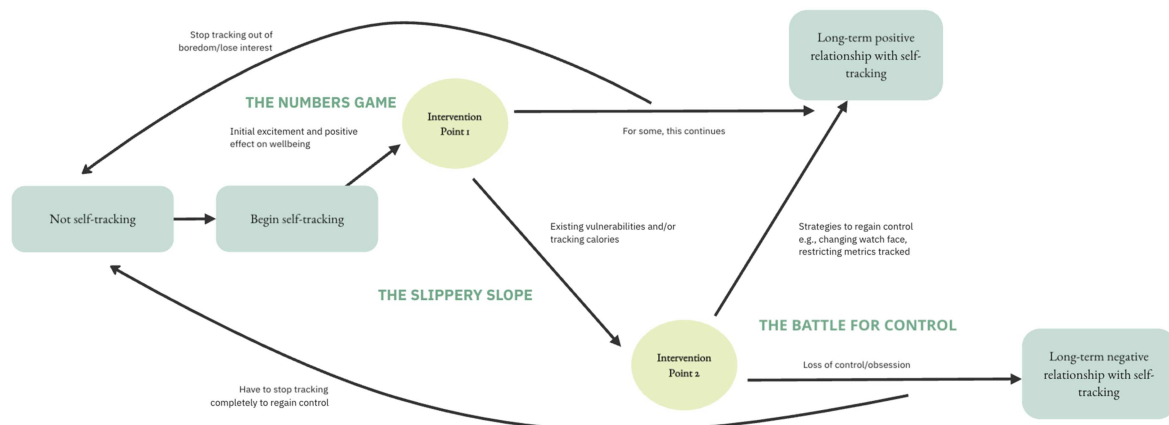


Figure 2. The changing relationship between adolescents' psychological wellbeing and self-tracking experiences.

illustrates the connections between the themes described in the Results (the numbers game, the slippery slope, and the battle for control) and serves as a starting point for further research.

The initial draw of self-tracking can be compelling and young people find comfort in seeing their data. For some, this initial excitement and positivity around self-tracking remains; however, others are more concerned about the long-term effects of self-tracking: *'in the short term I would say, uh, nice, but in the long run I wouldn't like say it's a good thing'* (Ophelia, 17). During these journeys, long-term tracking can have harmful effects on psychological wellbeing (Etkin, 2016) and, in line with the findings of Zimdars (2021), can lead to a feeling of disconnection from one's body. Adaline (16) explained one potential path for adolescents:

At the end of the day you get a big thing going, 'Well done, you've reached your goal'. That's going to like give you a good buzz and then you end up stuck with that and you end up with a situation where people can't cope without these apps and it gets more and more extreme until they're actually like a danger to themselves.

This narrative journey points to two key points that should be considered in relation to interventions that could support adolescents when they are navigating their own paths towards what they imagine to be positive engagement with self-tracking tools. The adolescents in this study showed that at the beginning of the tracking journey, seeing improvements can boost self-esteem and have positive effects on their psychological wellbeing. For some, this relationship continues to be positive. For others, particularly when calorie-tracking or existing vulnerabilities are involved, this relationship can start to take a darker turn (Chen et al., 2024).

The affordances of and motivations for self-tracking are not static (Epstein et al., 2015). Adolescents do not necessarily move linearly through the stages in the diagram shown in Figure 2. The same affordances of self-tracking tools will not always encourage or discourage the same experiences across individuals or time (Davis & Chouinard, 2016; Davis, 2020). This highlights the importance of moving away from a 'one size fits all' approach to adolescents' self-tracking (Freeman & Neff, 2023), instead providing continued support for adolescents throughout their individual self-tracking journeys and being careful not to make assumptions about their experience.

Surveying this journey, the first moment of note is the critical juncture between positive and negative experiences of tracking. How can we support adolescents who are engaging in calorie-tracking or tracking alongside existing challenges to promote positive health behaviors and protect them from *'unhealthy habits'* (Ophelia, 17)? Adolescents clearly distinguish calorie-tracking from other forms of self-tracking. How might we think about catching adolescents before they fall down this slippery slope so that we can set them up with the best possible chance of a long-term positive self-tracking experience?

The second important moment identified is the point after which adolescents feel they have lost control of their self-tracking. Whilst some adolescents in this research were keenly aware of the strategies that they could put in place to regain their self-management of these practices, this was not universal. Here, it is important to consider how we might bolster adolescents' feelings of autonomy and self-efficacy so that they can manage their own choices and behaviors and realize the benefits of self-tracking technologies to support both their physical and mental health.

Limitations and recommendations for future research

This study has several limitations. Firstly, while efforts were made in the recruitment approach to reach adolescents with varying experiences with self-tracking tools (e.g. advertising the project as a 'lifestyle' study), it is possible that self-selection bias led those with pre-existing interests in self-tracking, technology, or health and fitness to volunteer to take part. Furthermore, the interviews took place during various stages of the COVID-19 pandemic, including periods of lockdown in the UK. Whilst it is likely that the themes identified remain relevant beyond this particular context, and indeed are echoed in other research (Chen et al., 2024), further work is needed to uncover if and how these experiences might look different across contexts.

Akin to Honary et al.'s finding that 'obsessive engagement with [self-tracking] apps may reflect problematic use' (2019, p.14), adolescents described a threshold beyond which self-tracking might have

harmful effects on their psychological wellbeing. This threshold centered around obsession and a feeling of loss of control, often precipitated by a slippery slope of existing vulnerabilities or the tracking of calories (Chen et al., 2024). Future research might consider exploring this threshold; however, as self-tracking is often a continuous, physical practice, it is hard to capture frequency of use or similar constructs. To add additional complexity, in parallel with other research (Clark et al., 2022), many of the adolescents in this study spoke of internalized tracking even without engaging with tools. Researchers should consider the many guises of self-tracking and work to capture different types of engagement.

Echoing previous work (e.g. Honary et al., 2019), adolescents clearly articulated a difference between calorie-tracking and the tracking of other physical activity data. Future studies might consider asking more specifically about tracking calories to see if the relationship differs from what we have seen here when asking about general health and fitness tracking. Moreover, given the importance of age-appropriate resources to support adolescents' self-tracking (Freeman & Curtis, 2022, Freeman & Curtis, 2023), researchers should explore whether different resources required for calorie-tracking, as compared to tracking other data (e.g. step count), to protect adolescents' wellbeing when engaging with these tools.

Ultimately, future work should prioritize including young voices. Working closely with adolescents at every stage of research and technology design processes is essential for creating systems that feel empowering for and meet the needs of this important population.

Conclusion

Researchers, designers, and other stakeholders need to consider what is at stake here for adolescents and assess how we might reconfigure the emotional affordances of self-tracking tools to promote control and balance and provide a system that is designed with and for adolescent self-trackers to promote physical and mental health and wellbeing. Adolescents are able to clearly explain the situation as they see it before them, but they still have hope that change can happen. *'It doesn't necessarily have to be that way'*, Sawyer (16) said. They are optimistic about better self-tracking futures.

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